

B. Tech Degree V Semester Examination, November 2009**ME 503 MECHANICS OF MACHINERY**

(2006 Scheme)

Time : 3 Hours

Maximum Marks : 100

PART A

(8 x 5 = 40)

- I. (a) Explain two inversions of slider crank chain giving quick return mechanism.
- (b) State and prove Kennedy's Three Center's in Line Theorem.
- (c) Explain the terms, 'Tyre Synthesis and 'Dimensional Synthesis'.
- (d) Determine the Chebyshev spacing for the function $y = e^x$ for $0 \leq x \leq 4$, where three precision points are required.
- (e) Give a comparison between Involute and Cycloidal Gears.
- (f) Explain Reverted Gear Trains.
- (g) Give an account of different followers used in Cam construction.
- (h) Explain the laws of Friction.

PART B

(4 x 15 = 60)

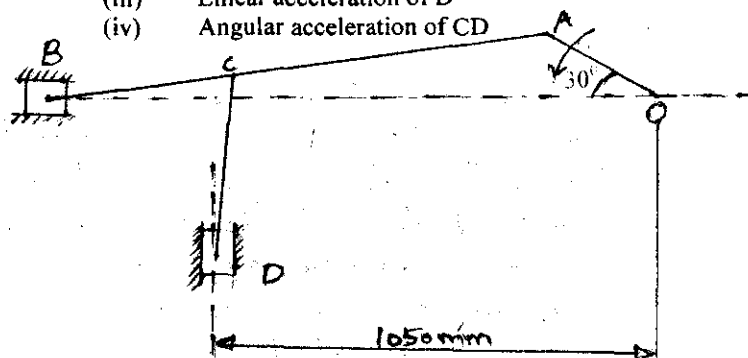
- II. (a) With the help of a figure explain steering gear mechanism. (5)
- (b) Explain the term Pantograph. Give its uses. (5)
- (c) The angle between the axes of two shafts connected by Hooke's joint is 18° . Determine the angle turned through by the driving shaft when the velocity ratio is maximum and unity. (5)

OR

- III. (a) Explain Coriolis' Component of acceleration. (5)
- (b) In the mechanism, as shown in the figure the crank OA rotates at 20 r.p.m anticlockwise and gives motion to the sliding blocks B and D. The dimensions of various links are OA = 300 mm, AB = 1200 mm, BC = 450 mm and CD = 450 mm. For the given configuration, determine :

- (i) Velocities of sliding at B and D
- (ii) Angular velocity of CD
- (iii) Linear acceleration of D
- (iv) Angular acceleration of CD

(10)



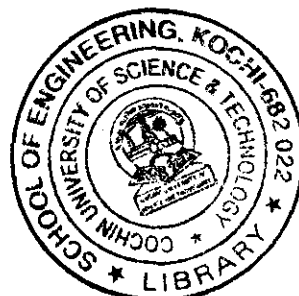
- IV. Design a four bar mechanism to co-ordinate the input and output angles as follows :

Input angles = $15^\circ, 30^\circ$ and 45°
 Output angles = $30^\circ, 40^\circ$ and 55°

(15)

OR

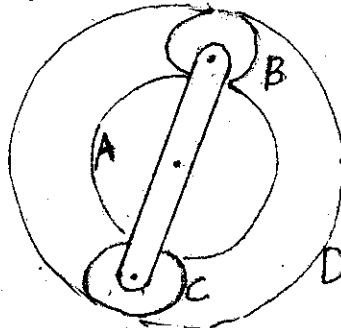
(Turn Over)



- V. Derive the Jacobian relation for positioning the manipulator having three axes. (15)
- VI. (a) State and prove the law of gearing. (5)
- (b) Two involute gears of 20° pressure angle are in mesh. The number of tooth on pinion is 20 and the gear ratio is 2. If the pitch expressed in module is 5 mm and the pitch line speed is 1.2m/s, assuming addendum as standard and equal to one module, find :
- (i) The angle turned through by pinion when one pair of tooth is in mesh
- (ii) The maximum velocity of sliding (10)

OR

- VII. (a) Explain how the gear trains are classified. Give at least one distinguishing feature of each type. (5)
- (b) An epicyclic train of gears is arranged as shown in the figure. How many revolutions does the arm, to which the pinions B and C are attached make.
- (i) When A makes one revolution clockwise and D makes half a revolution anticlockwise
- (ii) When A makes one revolution clockwise and D is stationary?
- The number of tooth on gears A and D are 40 and 90 respectively. (10)



- VIII. A cam is to be designed for a knife edge follower with the following data.
- (i) Cam lift = 40 mm during 90° of cam rotation with simple harmonic motion.
- (ii) Dwell for the next 30°
- (iii) During the next 60° of cam rotation, the follower returns to its original position with simple harmonic motion.
- (iv) Dwell during the remaining 180° .

Draw the profile of the cam when

- (i) the line of stroke of the follower passes through the axis of the cam shaft,
- (ii) the line of stroke is offset 20 mm from the axis of the cam shaft.

The radius of the base circle of the cam is 40 mm. Determine the maximum velocity and acceleration of the follower during its ascent and descent, if the cam rotates at 240 r.p.m. (15)

OR

- IX. (a) Derive an expression for the efficiency of screw jack considering collar friction. (5)
- (b) The pitch of 50mm mean diameter threaded screw of a screw jack is 12.5 mm. The coefficient of friction between the screw and the nut is 0.13. determine the torque required on the screw to raise a load of 25 kN, assuming the load to rotate with the screw. Determine the ratio of the torque required to raise the load to the torque required to lower the load and also the efficiency of the machine. (10)
